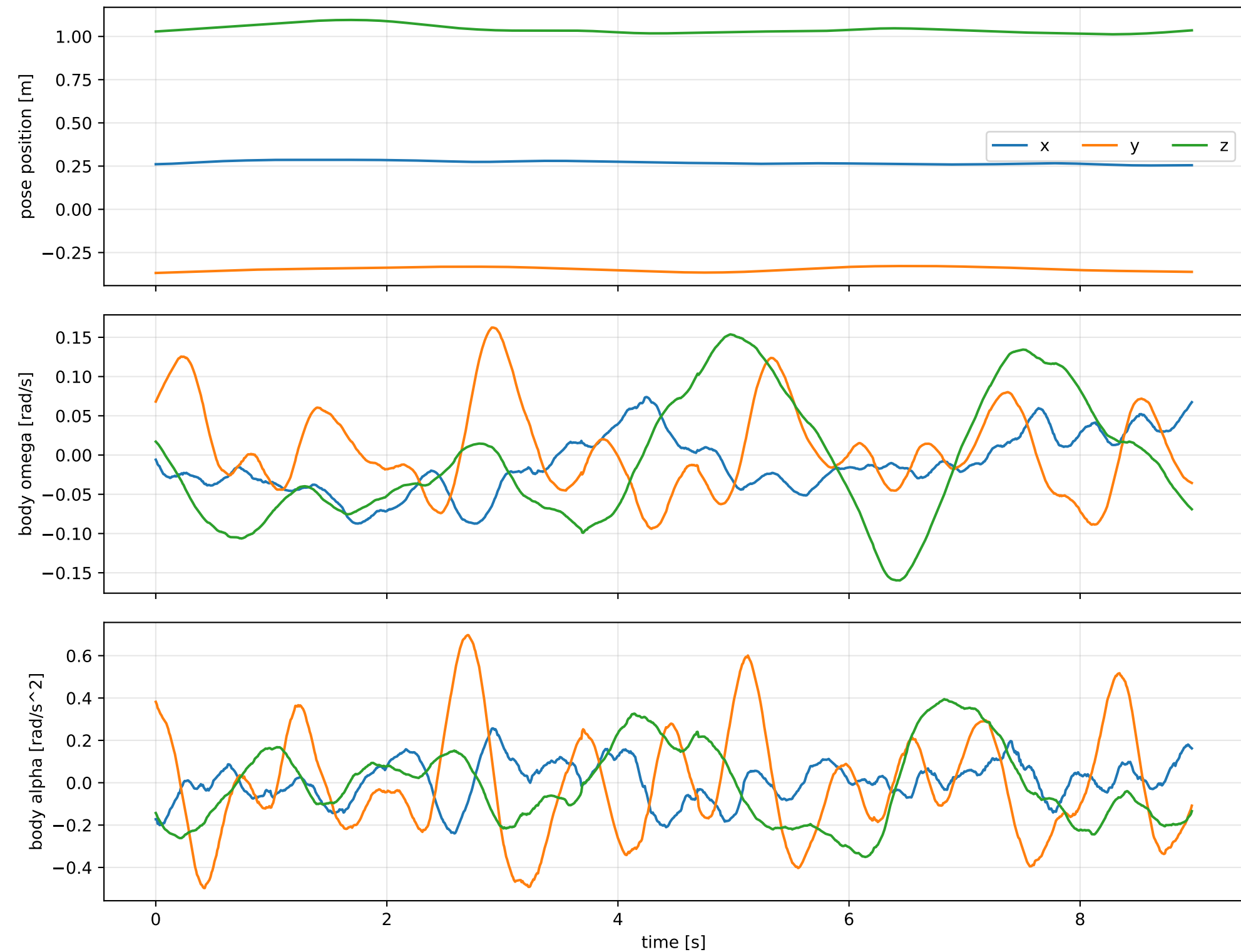
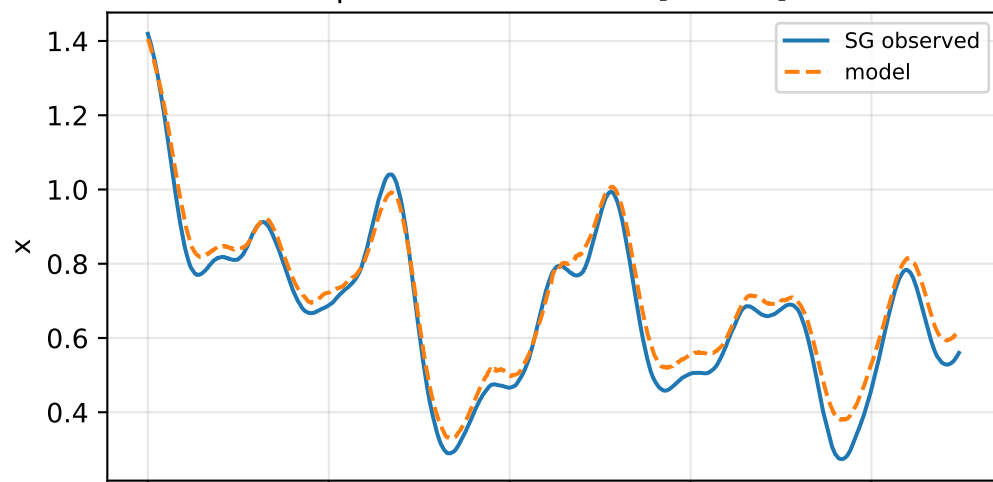
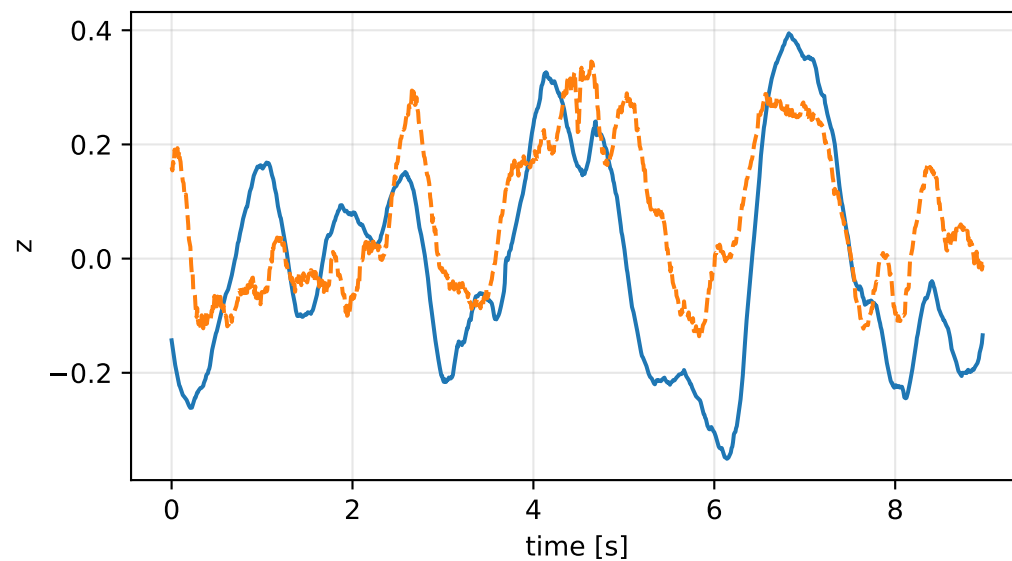
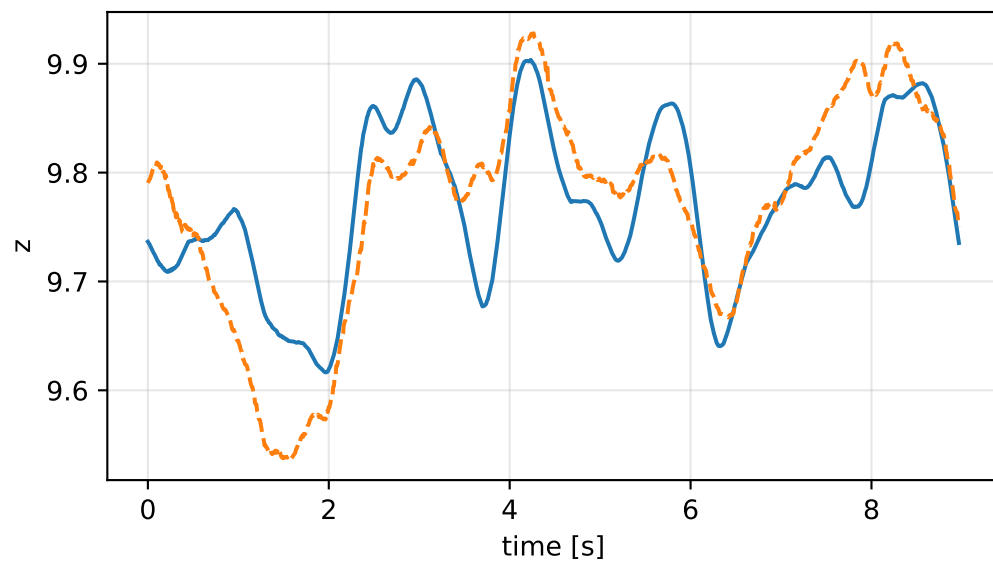
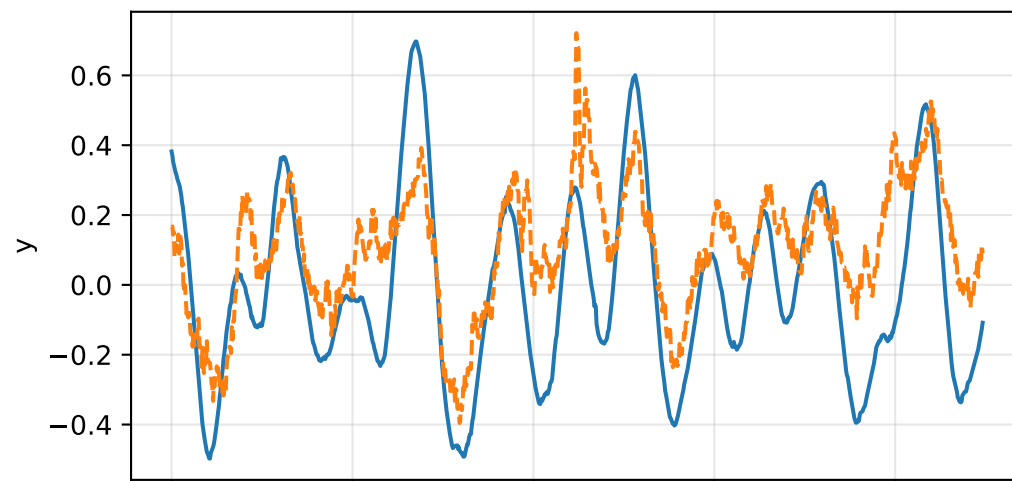
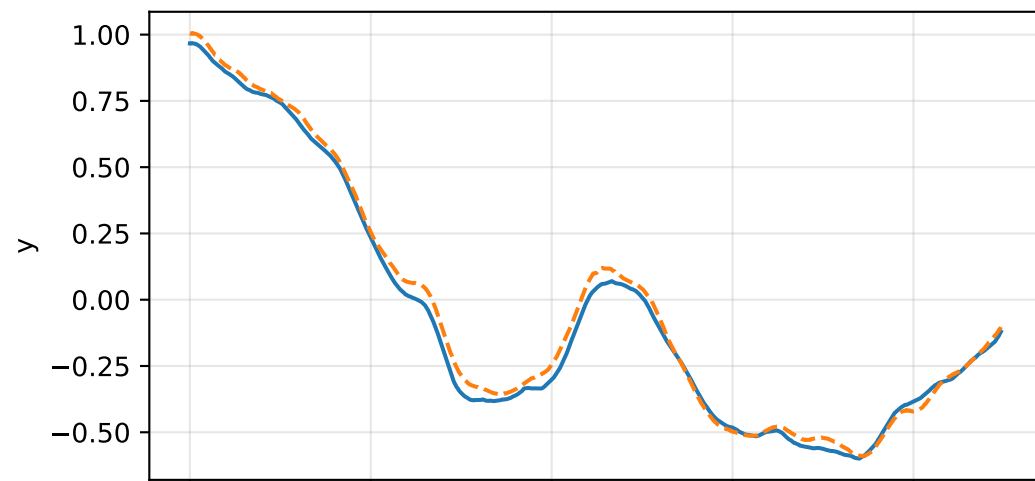
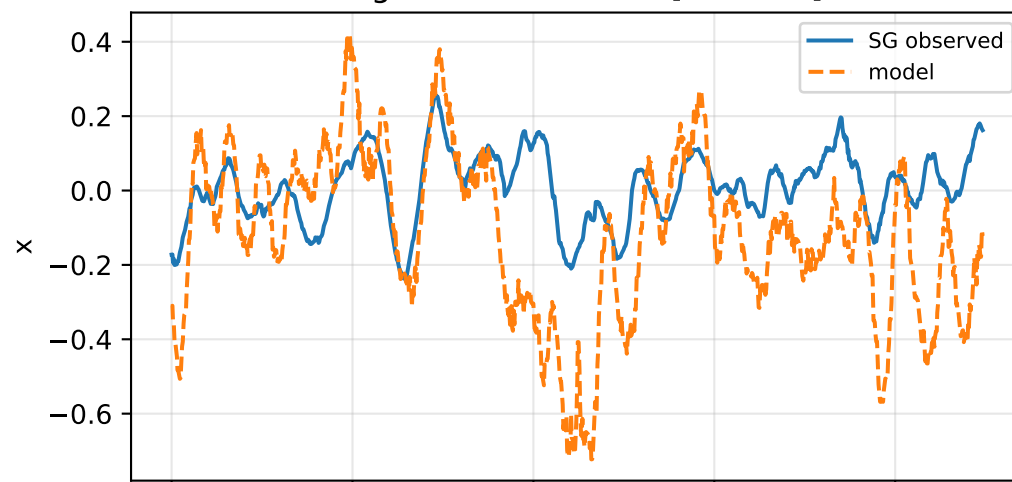


cov_full_no_position_rotation_cross: SG trajectory and derived motion

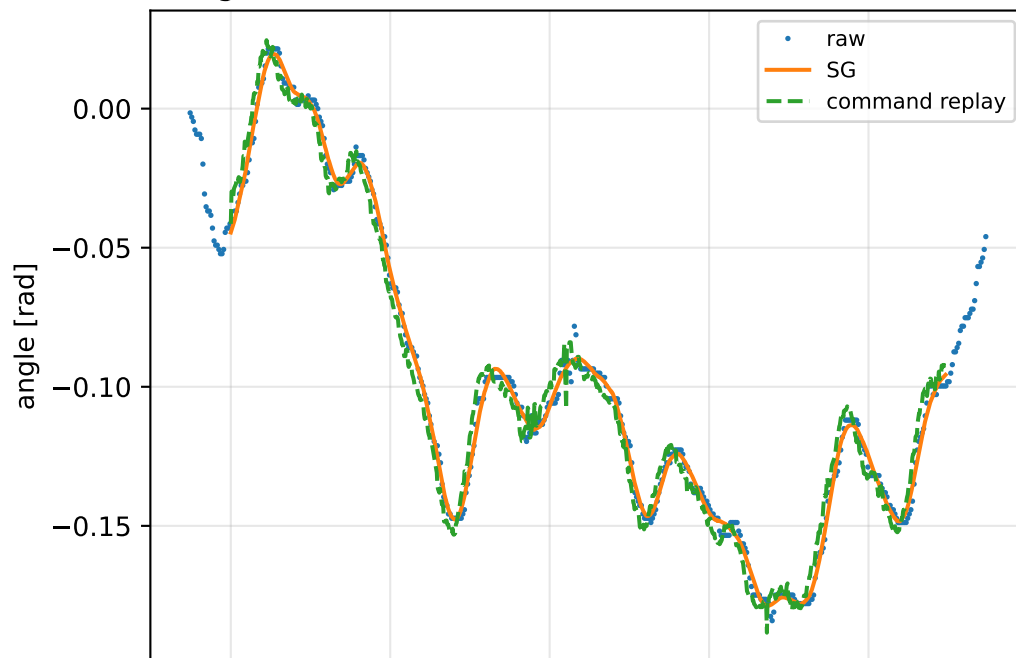


cov_full_no_position_rotation_cross: acceleration residual objective

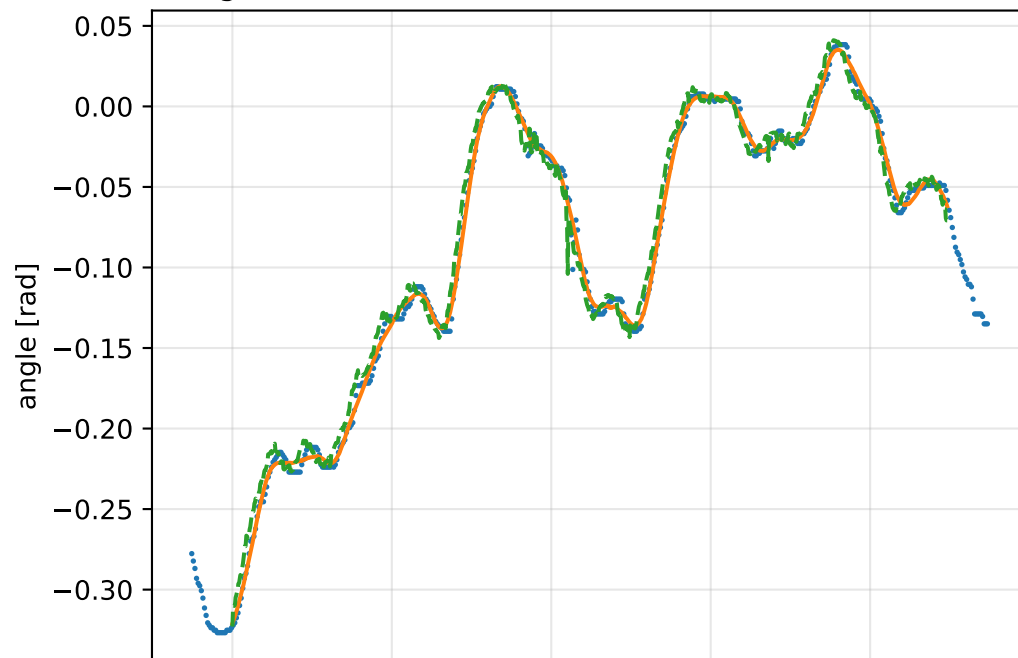
specific acceleration [m/s²]angular acceleration [rad/s²]

cov_full_no_position_rotation_cross: gimbal measurement audit (objective=measured_sg)

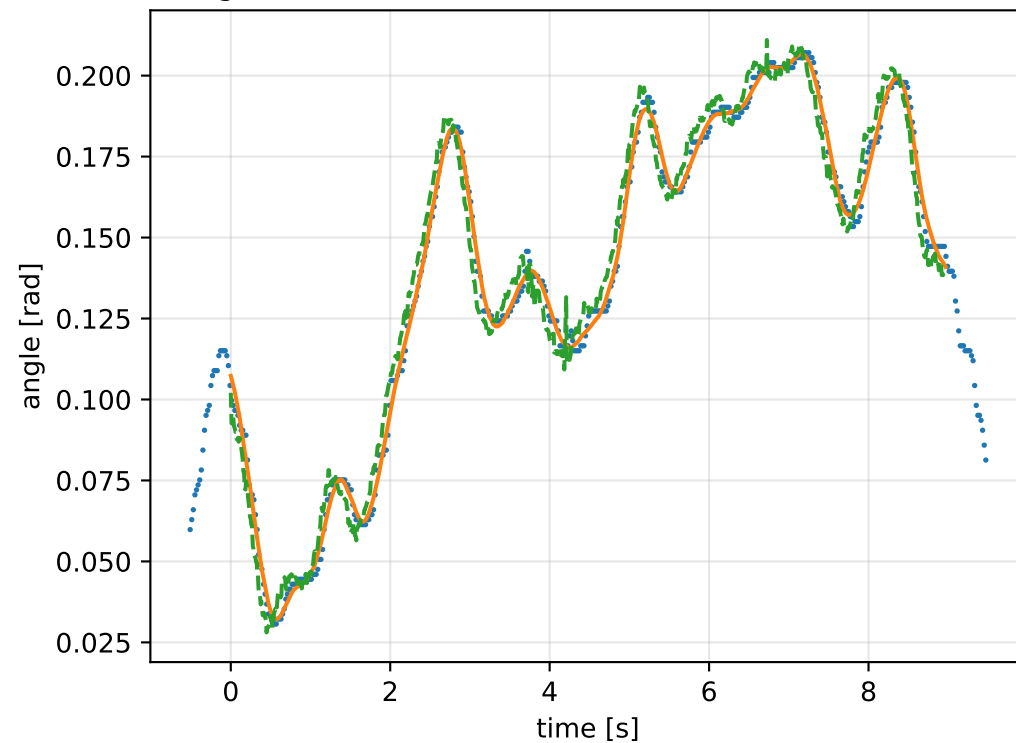
gimbal 1: RMSE=0.006345, max=0.01494 rad



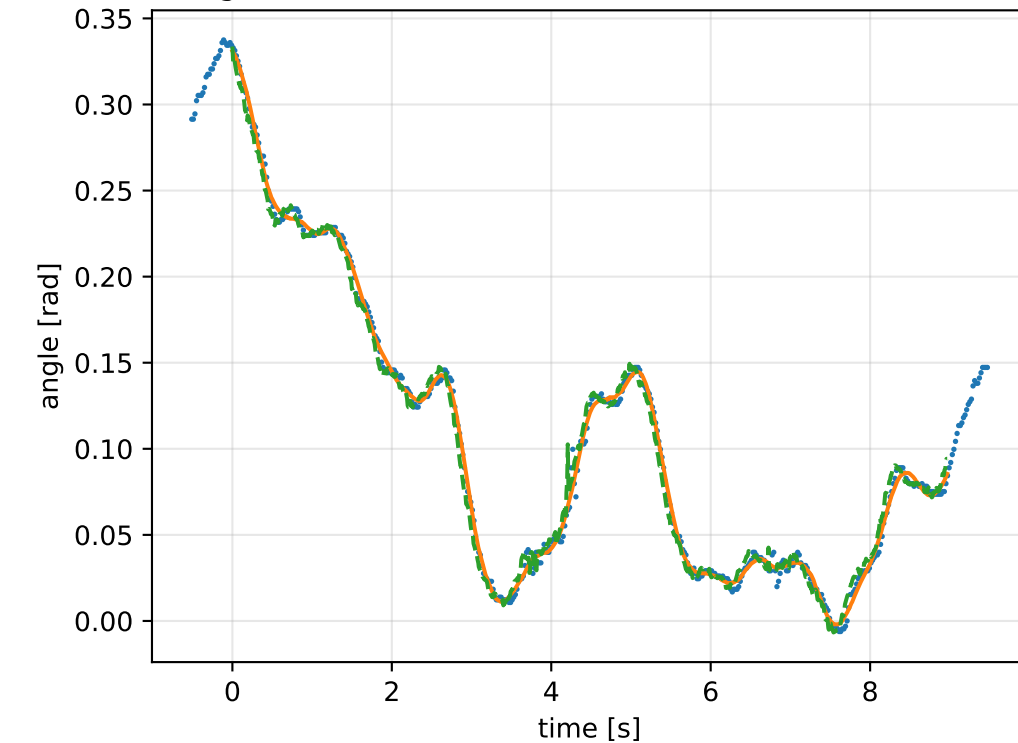
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

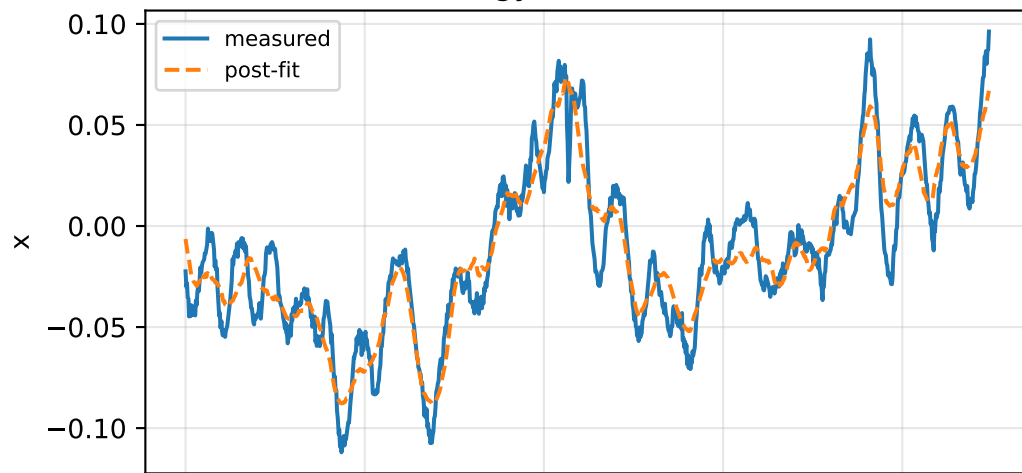


gimbal 4: RMSE=0.006321, max=0.03642 rad

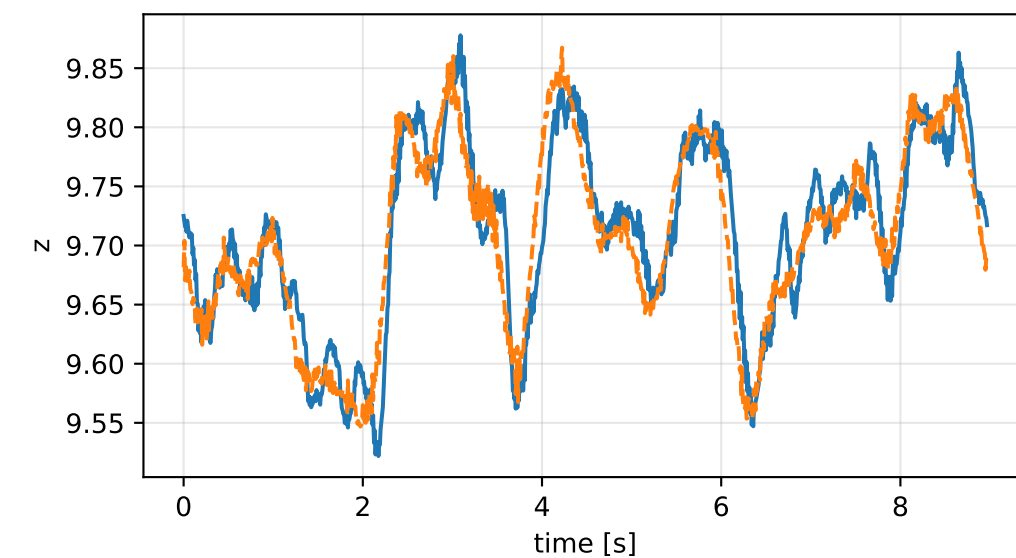
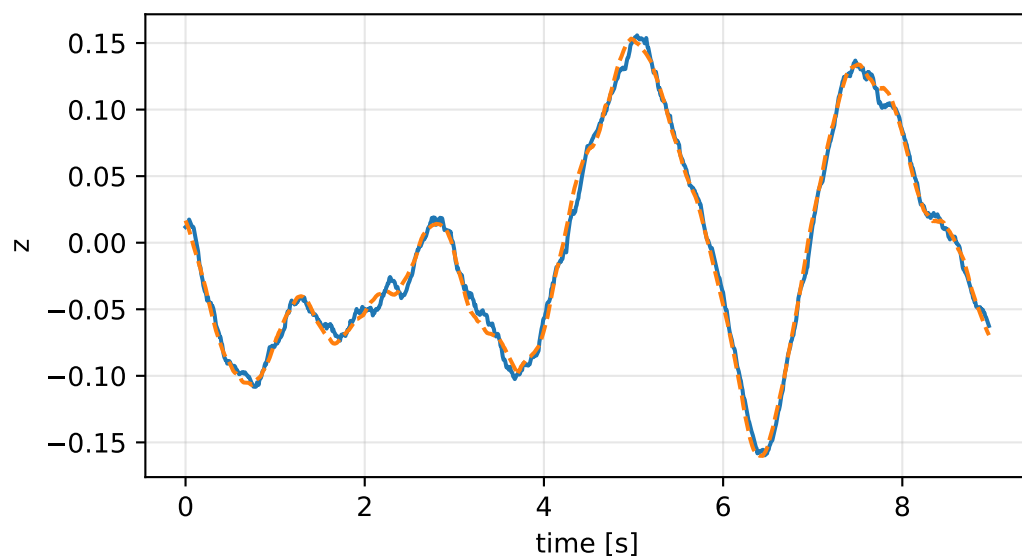
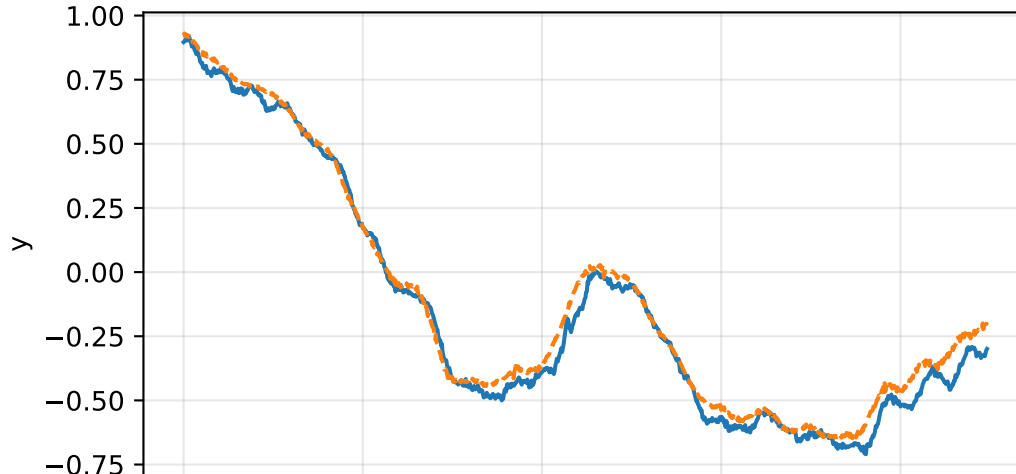
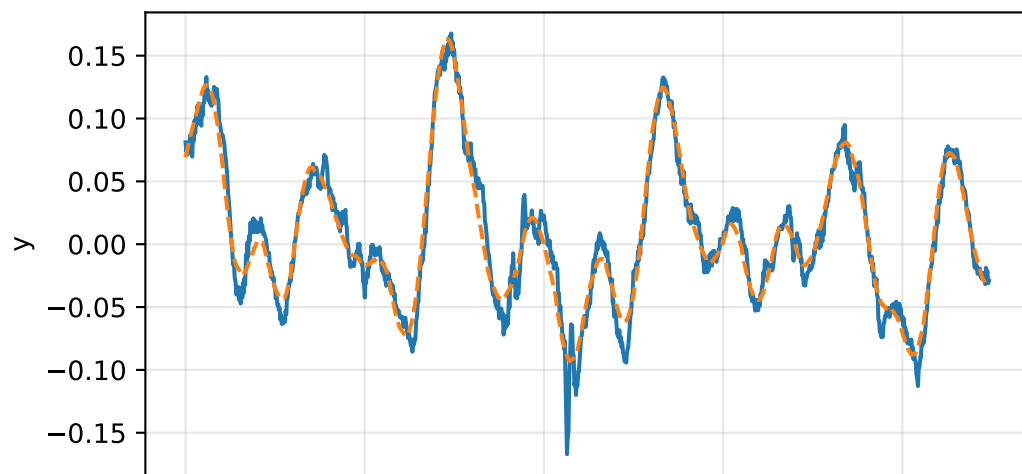
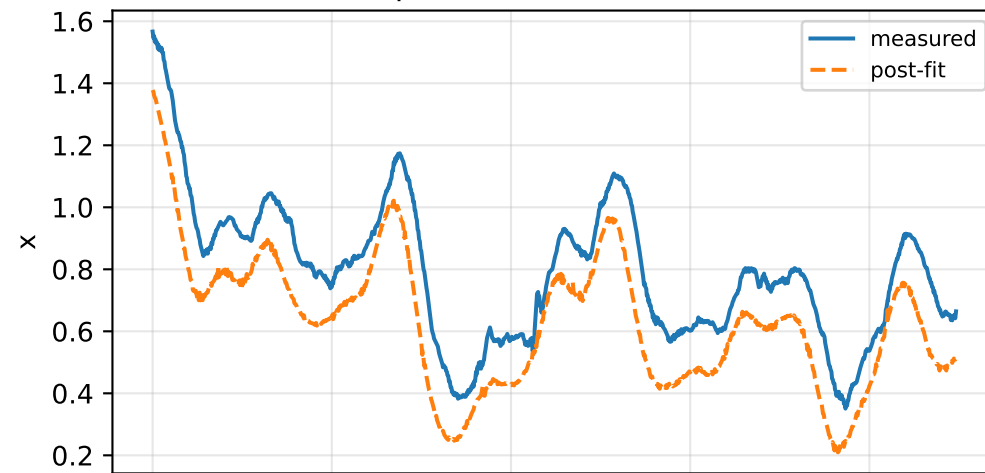


cov_full_no_position_rotation_cross: IMU comparison (diagnostic only)

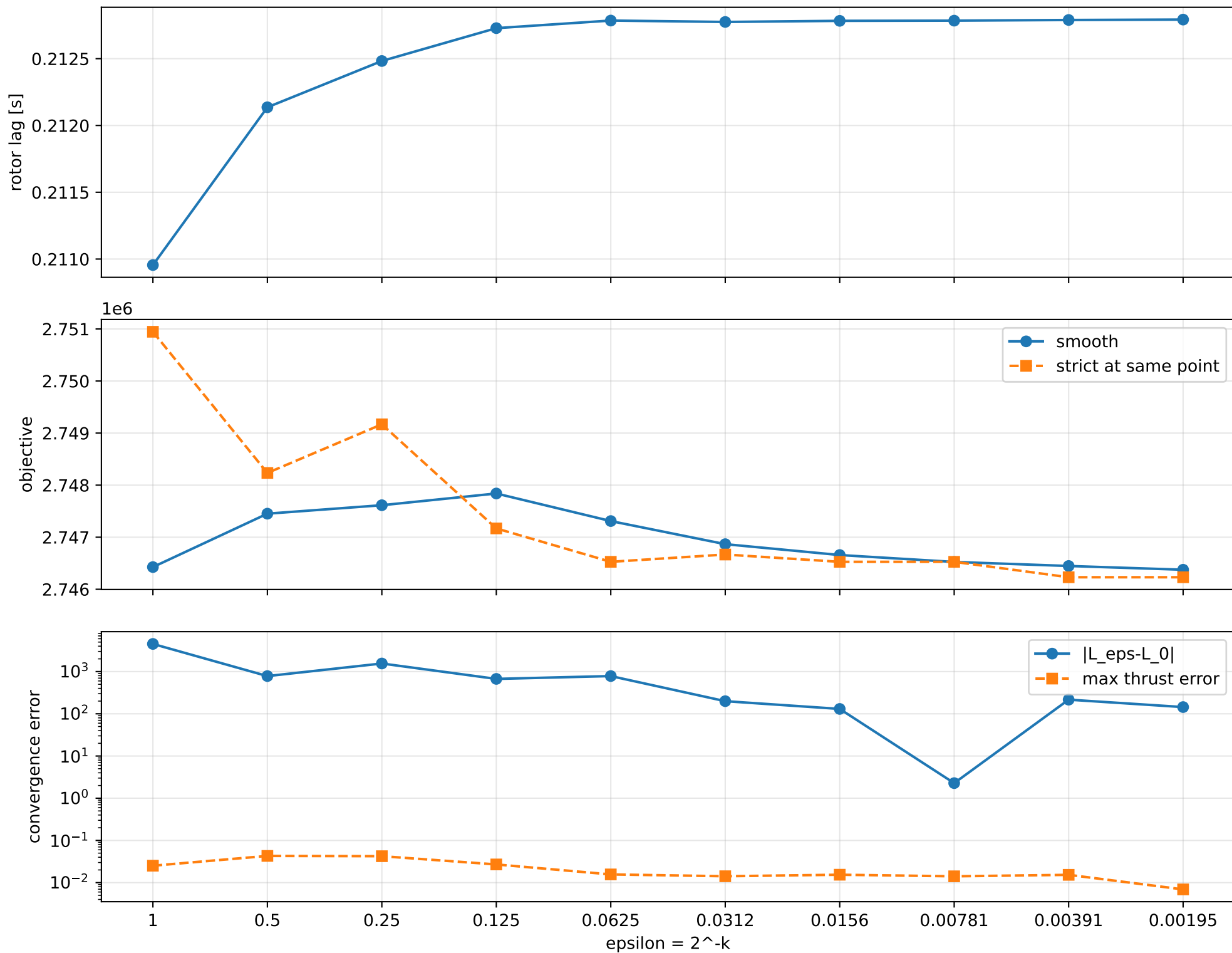
gyro [rad/s]



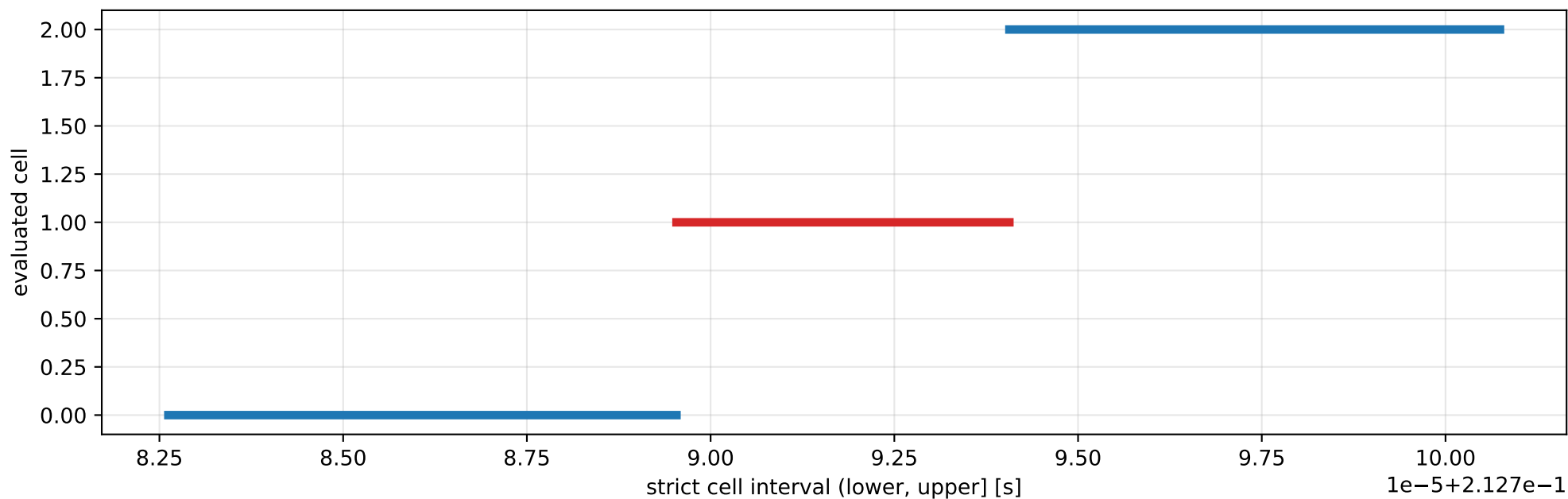
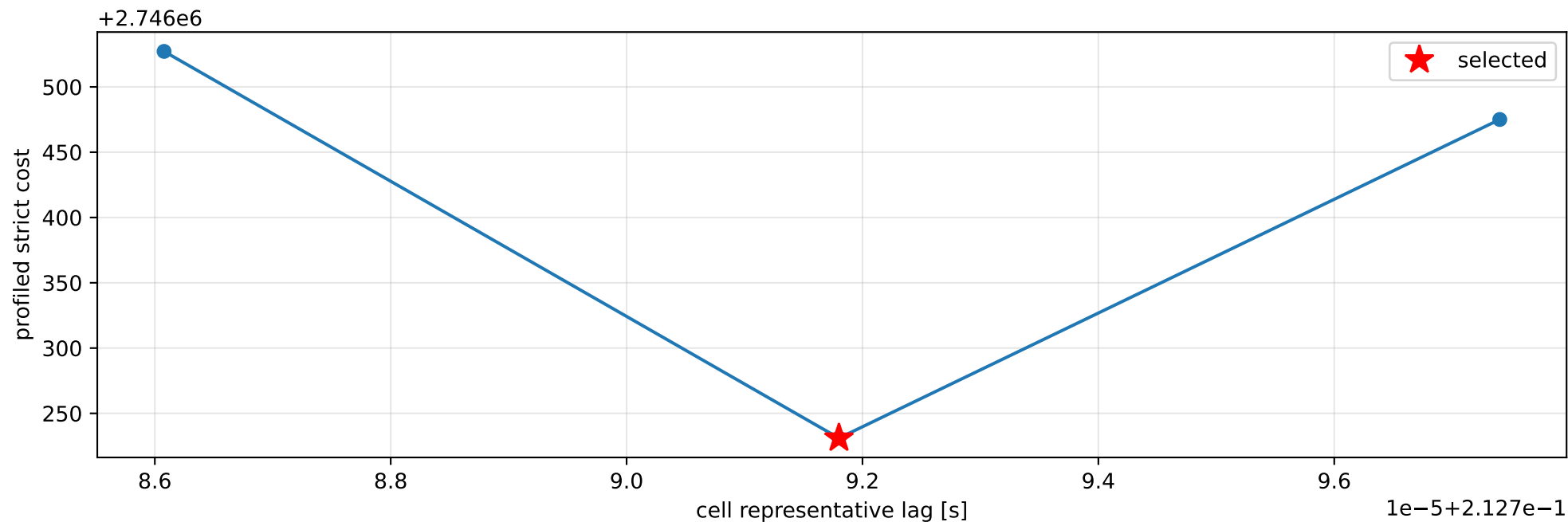
specific force [m/s^2]



cov_full_no_position_rotation_cross: smooth-to-strict continuation



cov_full_no_position_rotation_cross: exact strict-ZOH lag cells



selected (0.212789536, 0.212794065] s; final smooth 0.212792602 s; data boundary=False

cov_full_no_position_rotation_cross: parameter estimate

Case: cov_full_no_position_rotation_cross
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.72881085

inertia [kg m²]:

[[0.32988137 0.05992181 0.15516702]
 [0.05992181 0.53684247 0.05817804]
 [0.15516702 0.05817804 0.71155724]]

CoG body [m]: [-0.01828003 -0.06863143 0.02991213]

force effectiveness: [2.18745322 2.18656695 0.9300109 1.27024305]

scale-free J/m [m²]:

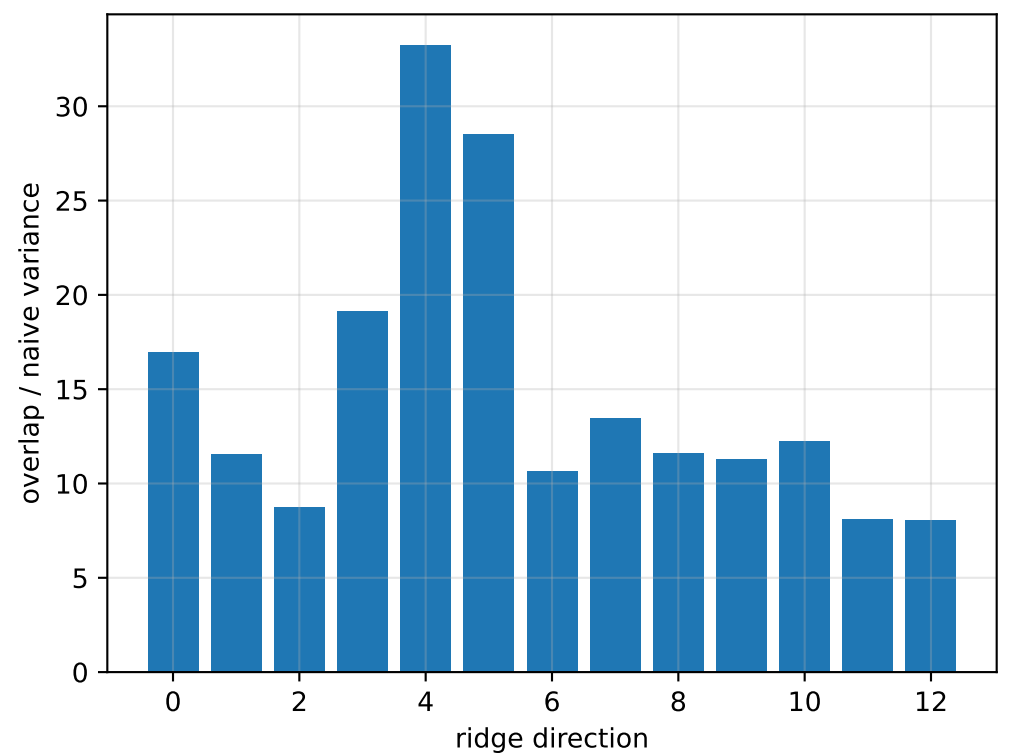
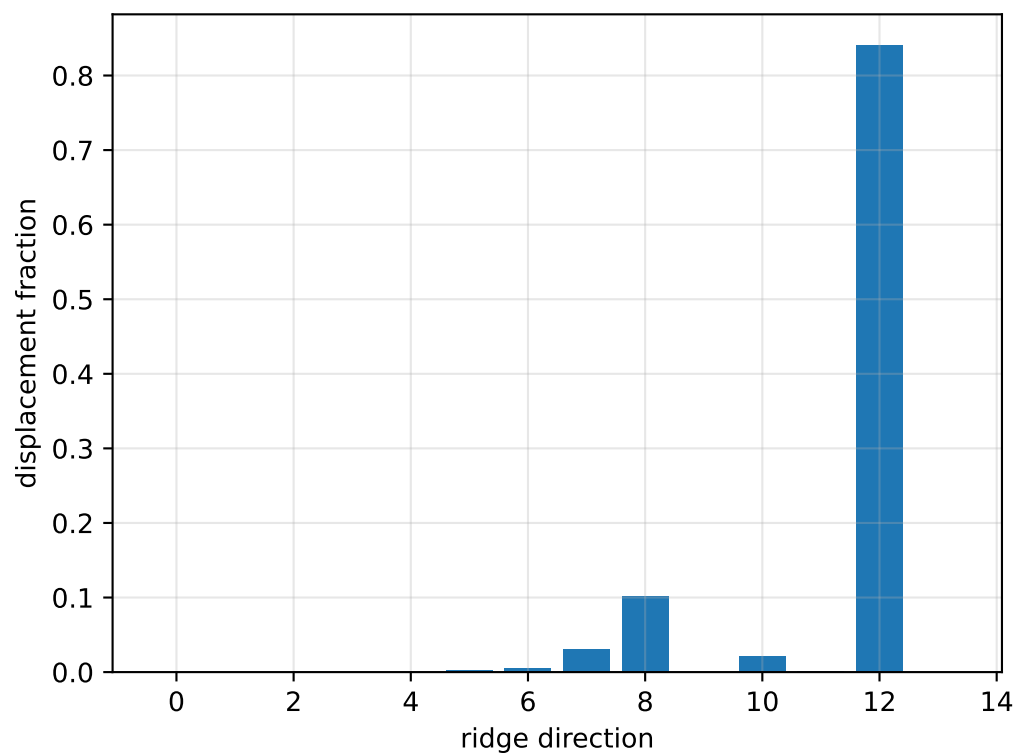
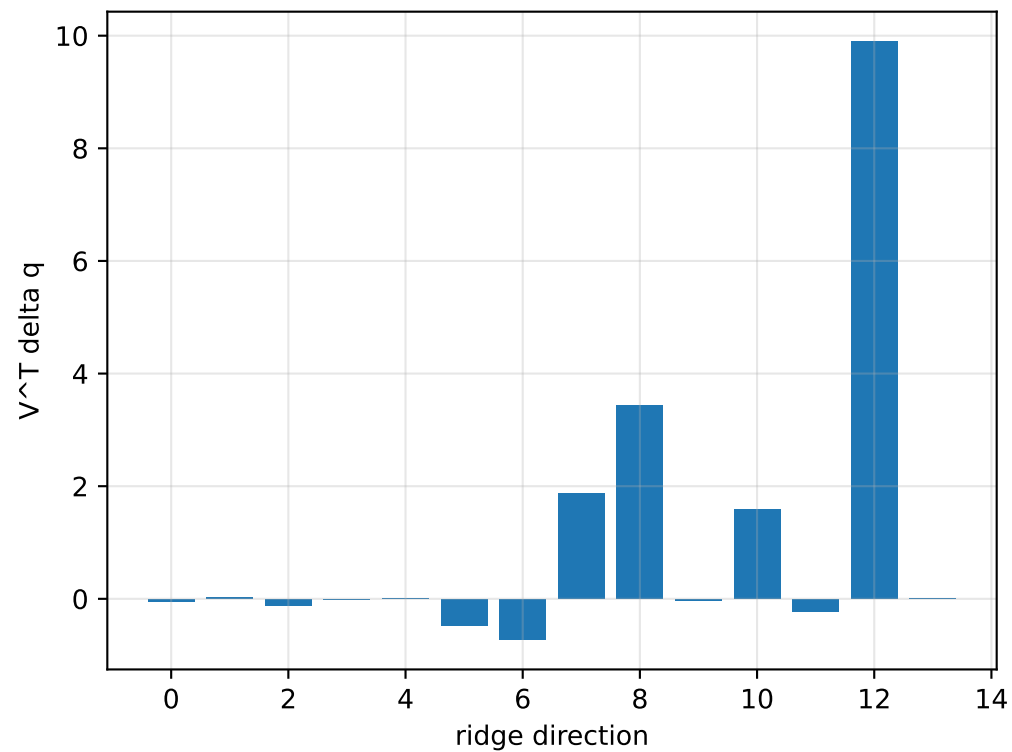
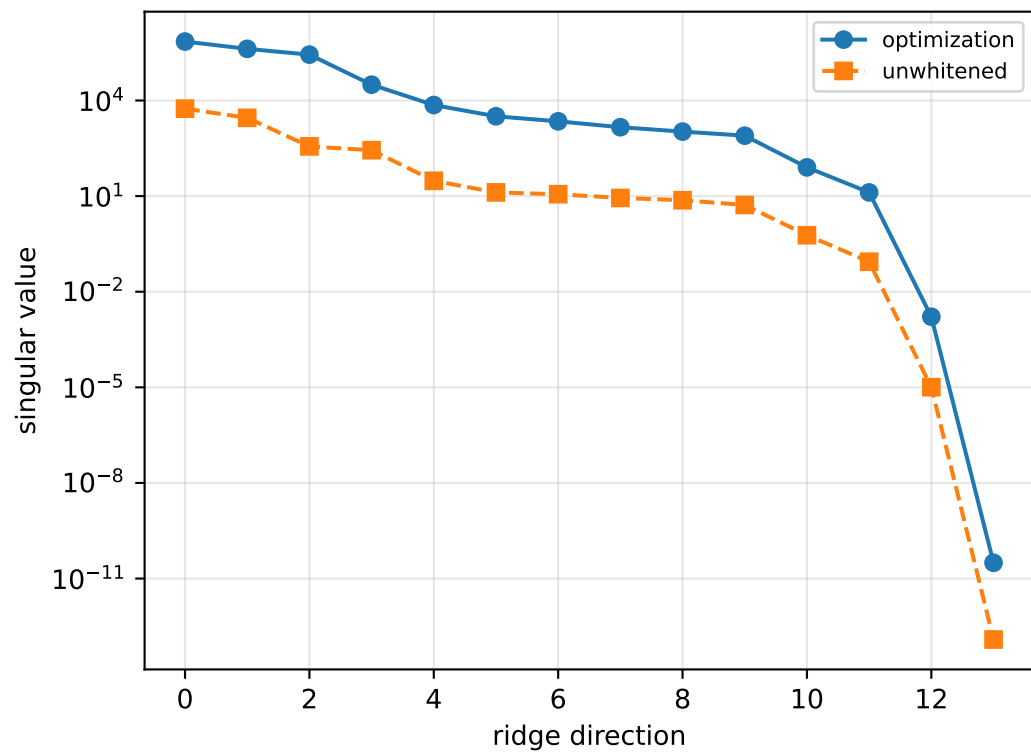
[[0.0697599 0.01267165 0.03281312]
 [0.01267165 0.11352589 0.01230289]
 [0.03281312 0.01230289 0.15047276]]

scale-free f/m: [0.46257998 0.46239256 0.19666908 0.26861786]

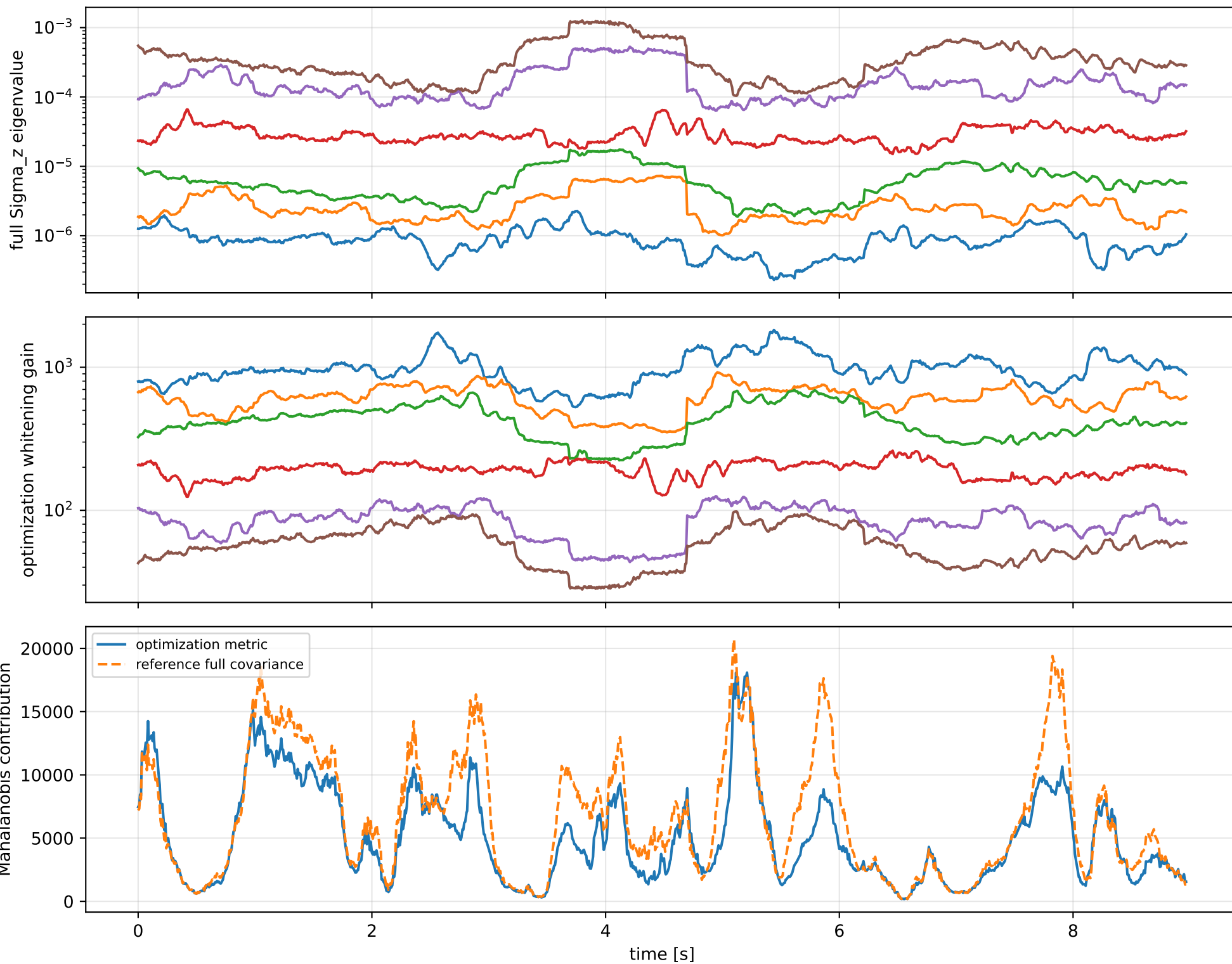
rotor lag cell: (0.212789536, 0.212794065] s

representative: 0.2127918 s

cov_full_no_position_rotation_cross: ridge spectrum (rank 13, nullity 1, ||jv||=5.67e-11)



cov_full_no_position_rotation_cross: optimization vs reference covariance



cov_full_no_position_rotation_cross: wrench / closure (force max 1.47e-14, torque max 1.11e-16)

